EXECUTIVE FIELD MANUAL · ISSUE #2

Zero-Downtime Database Architecture

How Staff+ Engineers evolve production schemas on 80M+ row tables without table locks or 504 timeouts.

- The ALTER TABLE trap: Why naive migrations exhaust connection pools
- The 3-phase Expand/Contract state transition blueprint
- Shadow verification & CDC-driven dual-writing
- Safe backfill worker batching without replication lag spikes

Additive Changes & Dual-Writes

Never mutate existing columns in-place.
Decouple storage expansion from read logic.

1. Add new column as NULLABLE (Zero exclusive lock duration).

2. Application starts Dual-Writing: Write to (Old + New).

3. Read path remains 100% on the Old column.

4. If deployment fails, instant zero-loss rollback is guaranteed.

Batched Async Backfill with WAL Throttli

Migrating millions of historical records
without starving OLTP write capacity.

1. Background worker iterates by Primary Key in bounded batches (5,000 rows).

2. Dynamic sleep intervals based on replica replication lag (<10ms target).

3. Shadow Reads: Compare Old vs New outputs asynchronously in memory.

4. Emit metric alerts on schema divergence before flipping the read switch.

PHASE 3: THE CONTRACT & CLEANUP

Cutover & Safe Schema Teardown

Finalizing the state transition and safely reclaiming database overhead.

1. Flip read traffic 100% to the New column via feature flag.

2. Remove Dual-Write logic from code (Write only to New).

3. Soak in production for 72 hours under peak traffic.

4. Drop old column during scheduled low-traffic window.